GENERAL ORGANIZATION OF THE CELL

Dr. R. KHAROUBI

I – Generalities

All living organisms are composed of cells, the elementary units of organization of living matter. The first scientist to propose the term "cell" was the Englishman Robert Hooke in 1665, to designate the units that make up living matter. Unicellular or multicellular organisms are made up of cells that share a similar basic metabolism. Cellular specialization, however, is accompanied by variations of the "typical cell," which leads to a diversity of cell shapes and specific metabolic pathways, adapted to diverse modes and environments of life.

II – Classification

Currently, the living world is divided into three major domains: **Eukaryotes, Eubacteria, and Archaea.**

A – Eukaryotes

The eukaryotic cell essentially contains:

- A nucleus**:** containing, in the form of DNA, the information necessary for maintaining species characteristics and for the synthesis of specific proteins. Within the nucleoplasm, it contains DNA in the form of chromatin and the nucleolus, and is separated from the other cellular components by the nuclear envelope. This envelope has two membranes: the outer membrane in contact with the cytoplasm, and the inner membrane separated from the chromatin by the lamina. A perinuclear space separates these two membranes. Nuclear pores within the envelope allow both endogenous and exogenous substances to transit between the nucleus and the cytoplasm in both directions.

- A plasma membrane**:** a lipid bilayer associated with transmembrane and peripheral proteins. The main role of the plasma membrane is to maintain the integrity of the cell. Its outer surface interacts with the extracellular environment, while its inner surface faces the intracellular medium. These two surfaces do not have the same structure or functions, a phenomenon known as **membrane asymmetry**. The plasma membrane contains receptors and may present microvilli and intercellular junctions.

- A cytoskeleton: composed of microtubules, microfilaments, and intermediate filaments.

- An endomembrane system: including the endoplasmic reticulum (ER), the Golgi apparatus, and lysosomes, which communicate via vesicles formed by budding from their membranes.

- The endoplasmic reticulum is a network of interconnected cavities bounded by a membrane. It may bear ribosomes, forming rough endoplasmic reticulum (RER), or be devoid of them, forming smooth endoplasmic reticulum (SER).

- The RER, participates in the synthesis of proteins destined for secretion and most membrane proteins. Free ribosomes translate mRNA into proteins destined for the cytosol or for organelles such as peroxisomes and mitochondria. The SER is involved in lipid synthesis.

- The Golgi apparatus consists of a stack of flattened sacs (dictyosomes) associated with vesicles. It modifies, sorts, packages, and distributes the products synthesized by the ER. It is involved in glycosylation and is the main distributor of newly synthesized membranes.

- Lysosomes are membrane-bound organelles containing hydrolases capable of degrading most endogenous and exogenous molecules.

- Mitochondria**:** bounded by a double membrane. The outer membrane is highly permeable, while the inner membrane, less permeable, presents invaginations called **cristae**, which extend into the mitochondrial matrix. The cristae contain the respiratory chain and the ATP-synthesizing machinery, whose hydrolysis releases part of the energy required for cellular function. The matrix contains mitochondrial DNA and the components necessary for protein synthesis.

B – Eubacteria

Eubacteria constitute a highly diverse group of unicellular microscopic organisms, with variable shapes but sharing a general plan of organization and several structural features. In general, the bacterial cell is surrounded by a resistant cell wall (Mainly composed of peptidoglycan). It is bounded by a plasma membrane composed of a lipid bilayer containing integral proteins. The bacterial chromosome (nucleoid) is arranged as a closed-loop filament occupying the center of the cell. It is not separated from the cytoplasm by an envelope. The cytoplasm is rich in proteins and contains ribosomes.

Eubacteria may also possess:

- a capsule of polysaccharidic nature,
- inclusions (glycogen, lipids, etc.), which serve as storage reserves,
- surface appendages such as pili and flagella.
- plasmid

C – Archaea

Archaea, formerly called archaebacteria, are unicellular microorganisms. They are very similar to eubacteria in terms of morphology and metabolism. With a size range similar to eubacteria (0.1–15 μm), archaea have a single circular chromosome, divide by binary fission, and often possess proteinaceous flagella for motility. Depending on the species, they may occur as coccoid, rod-shaped, or filamentous forms.

At the molecular level, archaea are much closer to eukaryotes than to eubacteria. For instance, many of their proteins—particularly those involved in information processing such as DNA replication, repair, and transcription—are far more similar to eukaryotic proteins than to bacterial ones. In numerous cases, archaea and eukaryotes share proteins absent from eubacteria.

However, archaea also possess unique characteristics distinguishing them from both other domains. Their envelopes do not contain peptidoglycan (present in most eubacteria) but are generally composed of a protein layer covering the plasma membrane. Moreover, archaeal phospholipids are entirely different: they are isoprenoid ethers (derived from a long-chain alcohol) rather than fatty acid esters as in eubacteria and eukaryotes.

Archaea are notable for the diversity and originality of the habitats they occupy. Some are well known for their ability to thrive in extreme conditions (anaerobic, high salinity, high temperature, or extreme depth) and colonize environments such as marshes, deep-sea hydrothermal vents, saline waters, and acidic hot springs. Nevertheless, many archaea live in

more common and varied habitats such as soils, lakes, oceans, or even within the intestines and lungs of mammals.

III – Viruses

Viruses are pathogenic agents that cannot reproduce outside a host cell. They are obligate parasites. Lacking cytoplasm and a true nucleus, they consist of nucleic acid (DNA or RNA) associated with proteins.

There are DNA viruses (e.g., hepatitis B virus) and RNA viruses (e.g., HIV, SARS-CoV-2/Covid-19).

A virus requires a host cell to multiply. Each virus consists of a genome (DNA or RNA) surrounded by a protein capsid that protects its genetic material. A virus has no metabolism of its own and cannot produce its proteins independently. Instead, it uses the host cell's machinery to synthesize its own proteins and replicate its genome. Indeed, the virus enters a host cell and hijacks its cellular machinery to replicate its genetic material and direct the synthesis of viral proteins. Subsequently, new viral particles are assembled, and the virions are released from the host cell.

Virions are released mainly through two mechanisms: **cell lysis and budding**. During cell lysis, the host cell is destroyed, releasing the newly formed virions. In contrast, during budding, enveloped viruses exit the cell by acquiring a portion of its plasma membrane, which forms their viral envelope. In this case, the host cell is not necessarily destroyed immediately and may continue to produce viruses for some time.

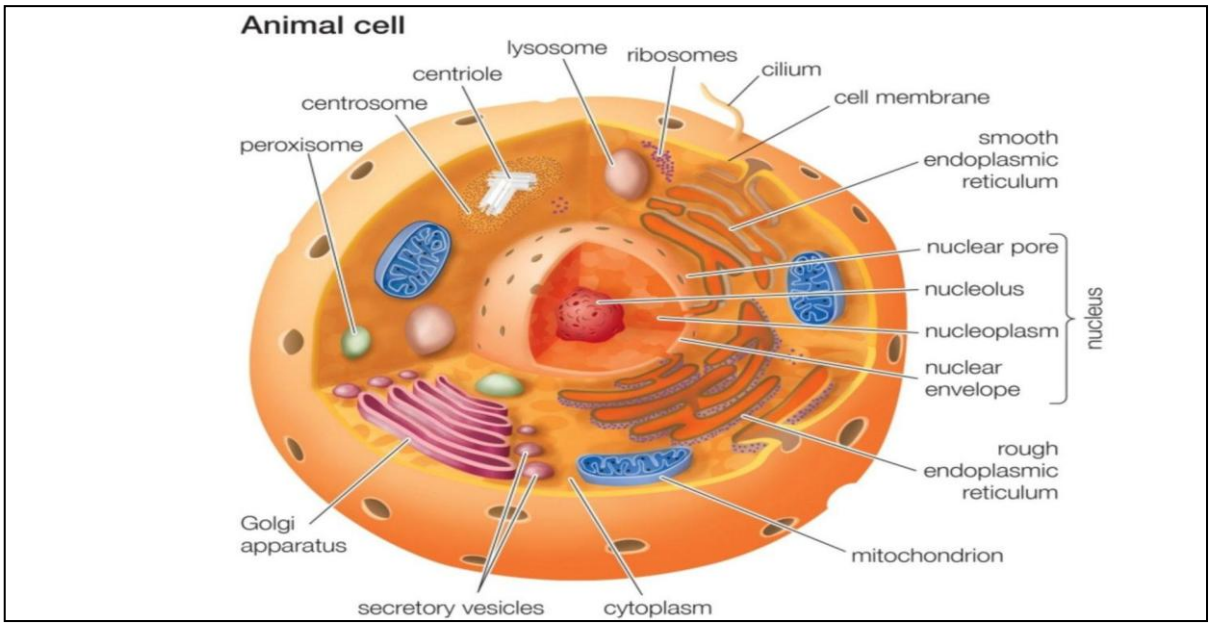


Figure 01. Organization of Animal cell

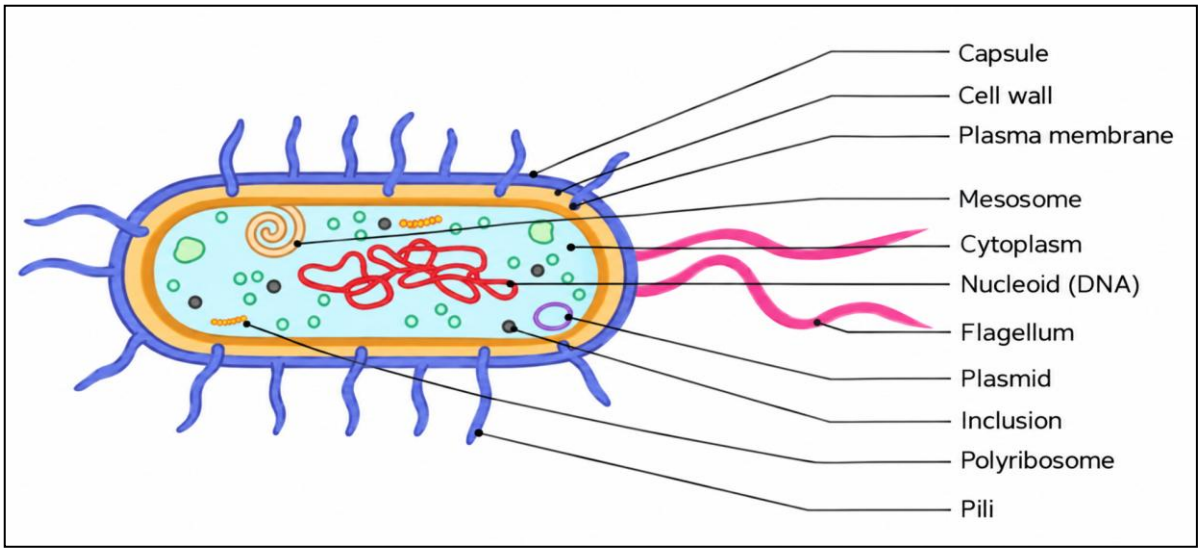


Figure 02. Ultrastructure of Bacteria

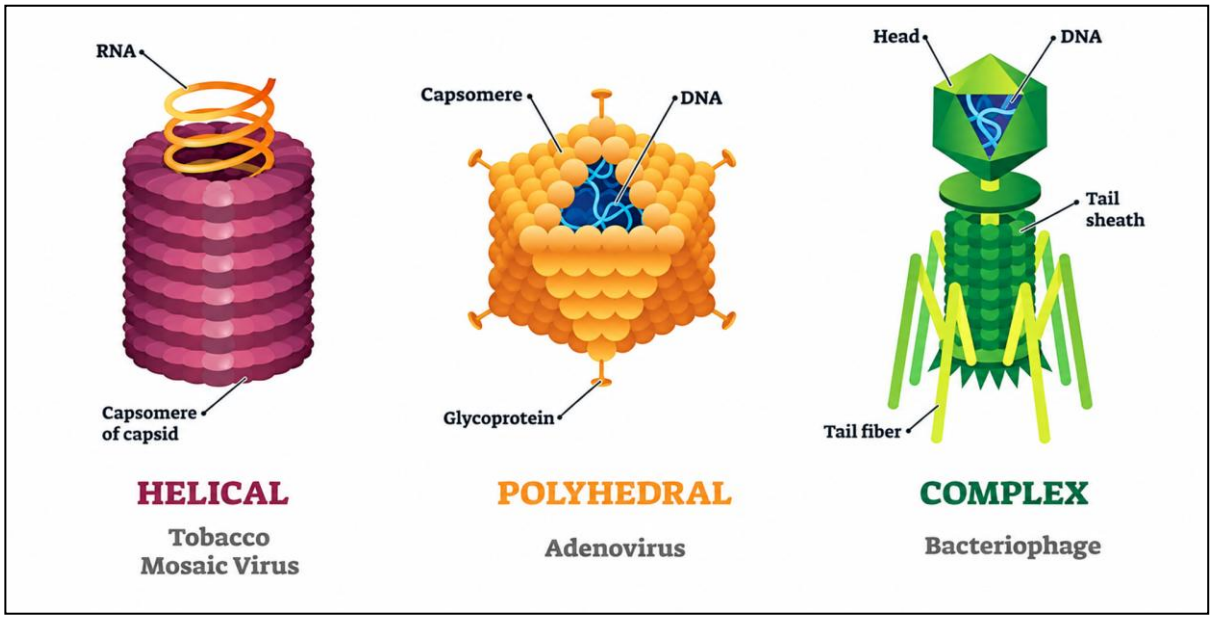


Figure 03. Organization of capsid